

Artifact Discrimination Framework for Nonequilibrium Electromagnetic Force Experiments

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Abstract

Small-force measurements in nonequilibrium electromagnetic systems are vulnerable to conventional artifacts including electrohydrodynamic flow, electrostatic attraction, Lorentz coupling, thermal drift, radio-frequency leakage, vibration, and nonlinear mechanical rectification. This paper develops an artifact-discrimination framework for evaluating reported anomalous force signals without assuming new physics. A general force-decomposition model is introduced, followed by a control matrix, a flowchart evaluation protocol, a directional “force compass,” and a set of illustrative effective-response bounds. The framework defines an effective susceptibility parameter, χ_{eff} , only after known artifact channels have been independently bounded. The goal is to establish a reproducible, falsifiable methodology for separating conventional force channels from candidate residual effects in precision electromagnetic experiments.

1 Introduction

Historical reports of thrust-like forces in electromagnetic systems include high-voltage asymmetric capacitors, electrohydrodynamic lifters, dynamic dielectric devices, RF resonant cavities, and Mach-effect thruster experiments. These systems share several properties: strong fields, geometric asymmetry, nonequilibrium drive conditions, and small measured forces near the sensitivity limit of the apparatus. Such conditions are also ideal for systematic error.

The central methodological problem is therefore not whether a measured force exists, but whether the measured force survives elimination of known mechanisms. The present work formalizes this problem in a conservative framework. A candidate residual force is treated as meaningful only after electrohydrodynamic, electrostatic, Lorentz, thermal, RF, vibrational, magnetic, buoyant, and nonlinear stick-slip effects have been experimentally bounded below the observed signal level.

This work does not propose a new force mechanism; rather, it defines the experimental conditions under which such a mechanism could be meaningfully identified.

2 General Force Decomposition

The measured force is decomposed as

$$\begin{aligned} F_{\text{measured}} = & F_{\text{EM}} + F_{\text{EHD}} + F_{\text{thermal}} + F_{\text{mechanical}} \\ & + F_{\text{electrical}} + F_{\text{magnetic}} + F_{\text{environmental}} + F_{\text{residual}}. \end{aligned} \tag{1}$$

Here F_{EM} includes Maxwell-stress and electrostatic forces, F_{EHD} includes ion-wind and corona-driven airflow, F_{thermal} includes thermal expansion and drift, $F_{\text{mechanical}}$ includes vibration and stick-slip rectification, $F_{\text{electrical}}$ includes cable forces and phase-dependent circuit coupling, F_{magnetic} includes Lorentz and material interactions, and $F_{\text{environmental}}$ includes buoyancy and convection. The residual term is defined only after each contributing channel has been independently bounded or experimentally constrained below the measurement threshold.

The effective residual susceptibility is then defined schematically as

$$\chi_{\text{eff}} = \frac{F_{\text{residual}}}{F_{\text{scale}}}, \quad (2)$$

where F_{scale} depends on the experimental class. For example, static dielectric systems may use an electric stress scale $F_{\text{scale}} \sim \epsilon E^2 A$, while RF cavities may use $F_{\text{scale}} \sim U/L$ with stored energy $U \sim QP/\omega$.

The parameter χ_{eff} is not assumed to represent a universal physical constant. Rather, it serves as a class-dependent normalization metric for comparing residual force magnitudes across different experimental configurations. Its interpretation depends explicitly on the chosen scaling relation and on the independently bounded artifact contributions.

3 Artifact Discrimination Matrix

Table 1 summarizes the core artifact classes, their typical signatures, and rejection tests. This table is intended as a minimum reporting standard for small-force electromagnetic experiments.

4 Flowchart Evaluation Protocol

Figure 1 presents the decision sequence used to classify a measured force. The protocol first removes environmental coupling, then tests electrical scaling, thermal response, phase/frequency dependence, and finally reversal or null geometries.

The residual force is then computed as

$$F_{\text{residual}} = F_{\text{measured}} - \sum_i F_{i,\text{bounded}}, \quad (3)$$

where each $F_{i,\text{bounded}}$ represents a measured or experimentally constrained artifact channel.

5 Directional Artifact Mapping: Force Compass

A useful diagnostic is not only the magnitude of a force but also its direction relative to the apparatus, fields, wiring, thermal gradients, and support geometry. Figure 2 gives a directional “force compass” for mapping candidate signals onto likely artifact axes.

6 Illustrative Effective Susceptibility Bounds

Table 2 gives an order-of-magnitude classification of several experiment families reviewed in the associated EVRT literature survey. The values are not universal constants; they are class-dependent response estimates or targets. Their purpose is to compare how large a residual response would need to be if the observed force were not explained by known mechanisms.

Table 1: Artifact identification and rejection criteria for nonequilibrium electromagnetic force experiments.

Artifact class	Typical signature	Rejection method
EHD / ion wind	Force in air, ozone smell, visible airflow, strong pressure dependence	Vacuum chamber, pressure sweep, smoke/flow visualization, polarity and gap tests
Electrostatic attraction	V^2 scaling, strong dependence on nearby conductors and grounding	Grounded shielding, distance sweep, dummy electrode geometry, conductor-position reversal
Lorentz / cable forces	Current-direction dependence, sensitivity to wire routing	Twisted pairs, coaxial feed, current reversal, dummy load, independent cable supports
Thermal drift	Slow response, hysteresis, relaxation after power-off	Thermal sensors, matched heater control, time-constant fitting, infrared monitoring
RF leakage	Orientation-sensitive signal, coupling to chamber or cables	RF shielding, absorber load, null cavity, field probes, external antenna survey
Vibration	Frequency-locked motion, mechanical resonance peaks	Accelerometers, isolation table, off-resonance drive, mechanical dummy load
Stick-slip rectification	Net bias from oscillatory drive, phase/frequency dependence, mount sensitivity	Frictionless suspension, mount reversal, symmetric mounting, shaker-only control
Magnetic coupling	Sensitivity to ferromagnetic components or magnetic damping	Magnetometer survey, nonmagnetic hardware, coil-current reversal, magnetic shielding
Buoyancy / convection	Pressure and temperature dependence, delayed drift	Vacuum test, thermal stabilization, enclosure control, pressure sweep
Residual candidate	Survives all controls with reproducible scaling	Quantify F_{residual} and report χ_{eff} with uncertainty

Artifact Discrimination Flowchart

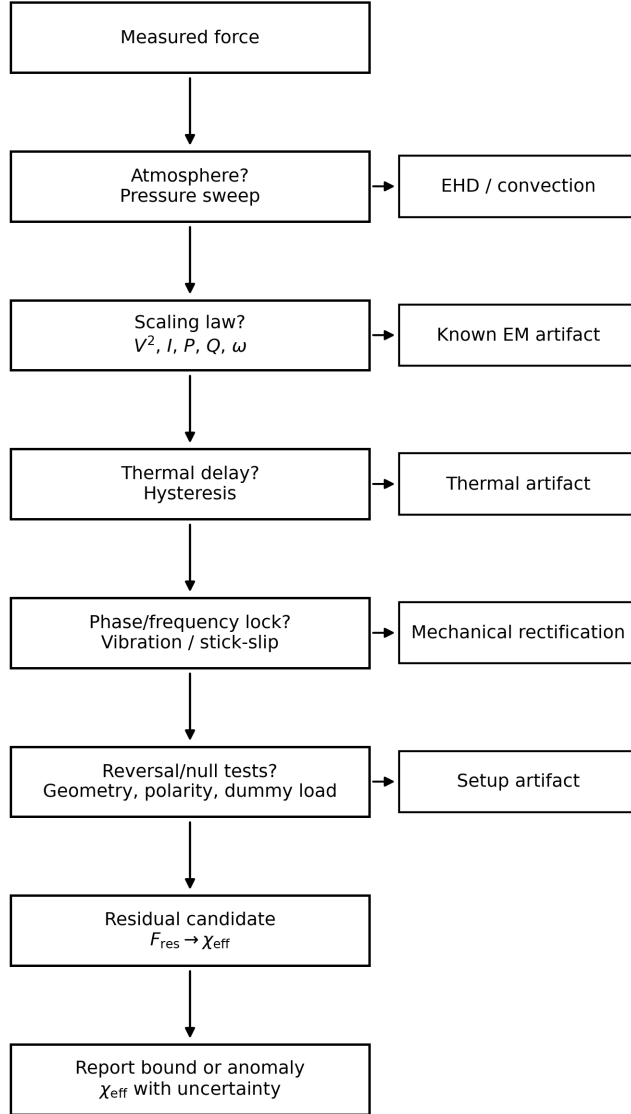


Figure 1: Artifact-discrimination flowchart. A candidate residual should only be assigned after the measured force survives pressure, scaling, thermal, phase, and reversal/null tests, and is subsequently reported through the χ_{eff} normalization.

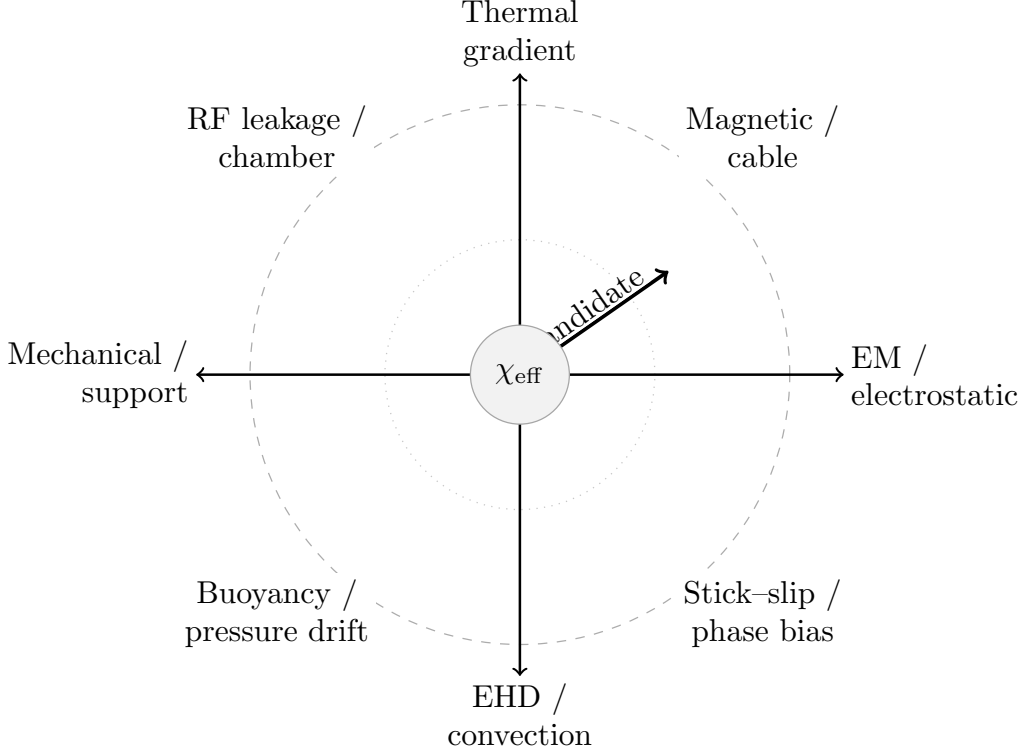


Figure 2: Force compass for directional artifact mapping. Primary artifact axes are shown as vectors, while secondary artifact families are placed as quadrant labels to avoid crowding. A true residual should not simply align with known artifact axes such as thermal gradients, cable routing, EHD flow, or mechanical support asymmetry.

Table 2: Illustrative χ_{eff} mapping for experiment families. Values are illustrative order-of-magnitude estimates for constraint comparison, not final measured constants.

Experiment family	Force scale	Dominant known channel	EVRT interpretation
EHD lifters / Biefeld–Brown in air	mg–g	Ion wind, corona, neutral-air momentum transfer	Residual should vanish in vacuum
Static high-voltage capacitors	mg	Electrostatic stress, EHD, leakage, grounding	Test E^2 scaling and shielding
RF resonant cavities	μN	Thermal drift, RF leakage, cable forces, Lorentz coupling	Test Q , P , ω , and null-load scaling
Mach-effect thrusters	μN	Vibration, thermal expansion, phase error, stick-slip	Test reversal geometry and symmetric null stack
Casimir / vacuum boundary forces	nN– μN	Established vacuum-force baseline	Useful lower-scale reference, not propulsion proof

Figure 3 plots an illustrative mapping of χ_{eff} against an effective electromagnetic drive scale or energy-density proxy. The plot emphasizes that different experimental families require different normalization scales and should not be collapsed into a single universal parameter without qualification. The shaded region represents an artifact-dominated comparison regime in which conventional mechanisms are expected to account for observed forces unless independently excluded. A horizontal reference line corresponding to an illustrative instrument sensitivity or target threshold defines a lower bound below which residual signals cannot be meaningfully distinguished from measurement noise.

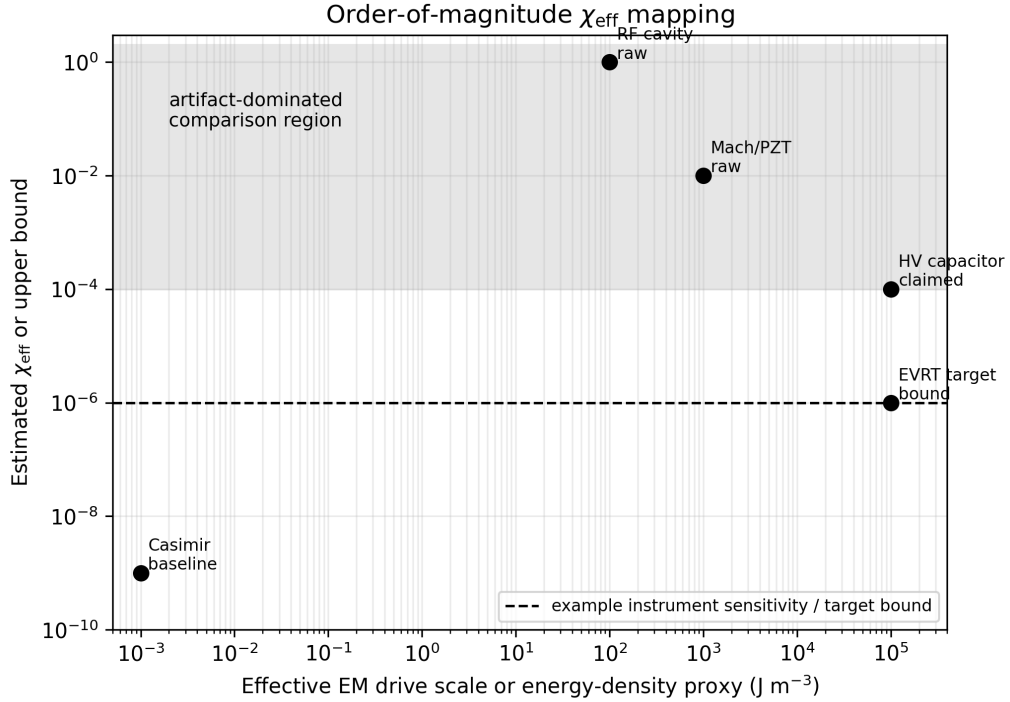


Figure 3: Illustrative order-of-magnitude χ_{eff} mapping versus an effective electromagnetic drive scale. Markers represent experiment families, not finalized universal constants.

7 RF Scaling Example

For an RF cavity, one simple residual-force scale is

$$F_{\text{res}} \sim \chi_{\text{eff}} \frac{U}{L}, \quad U \sim \frac{QP}{\omega}, \quad (4)$$

where Q is the cavity quality factor, P is the input power, $\omega = 2\pi f$ is the angular frequency, and L is the characteristic cavity length. This is not a proof of a force law; it is a normalization convention for comparing residual claims.

Figures 4 and 5 show the resulting scaling. At fixed frequency, the force scale grows linearly with Q . At fixed Q , the force scale decreases with increasing frequency for a fixed input power if the stored energy is approximated by QP/ω .

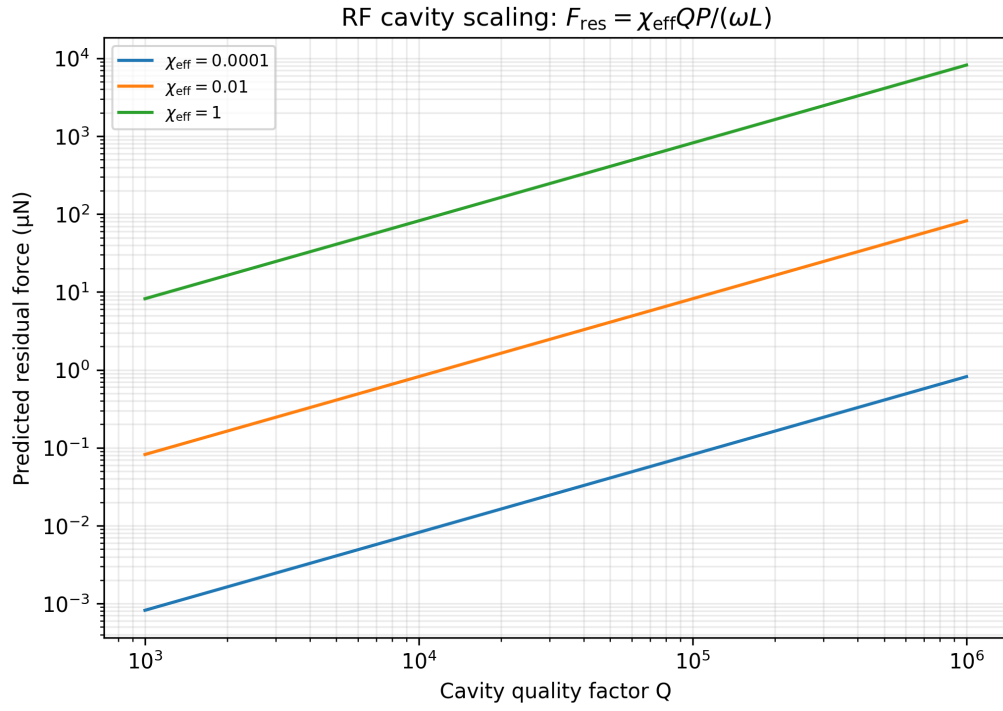


Figure 4: RF residual-force scaling versus cavity quality factor using $F_{\text{res}} = \chi_{\text{eff}} Q P / (\omega L)$ for fixed power, frequency, and length.

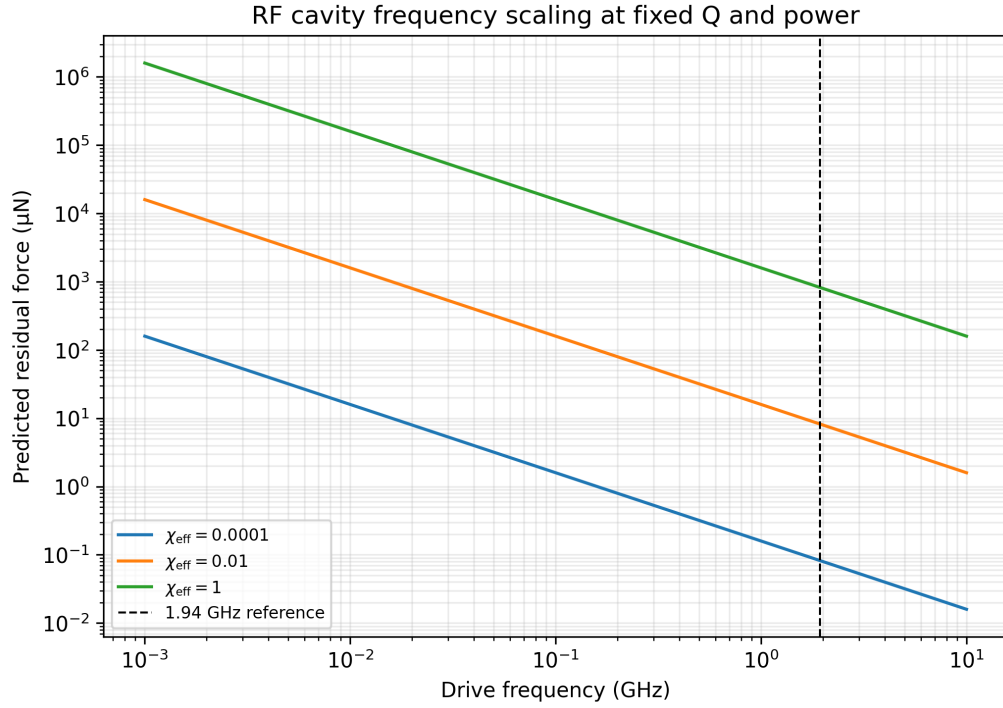


Figure 5: RF residual-force scaling versus drive frequency at fixed Q , power, and cavity length.

8 Torsion-Balance Implementation

Figure 6 maps a tabletop torsion-balance implementation onto the artifact-discrimination framework. The key idea is to couple any force readout to independent controls: pressure sweep, thermal monitoring, accelerometer readout, current reversal, and dummy-load operation.

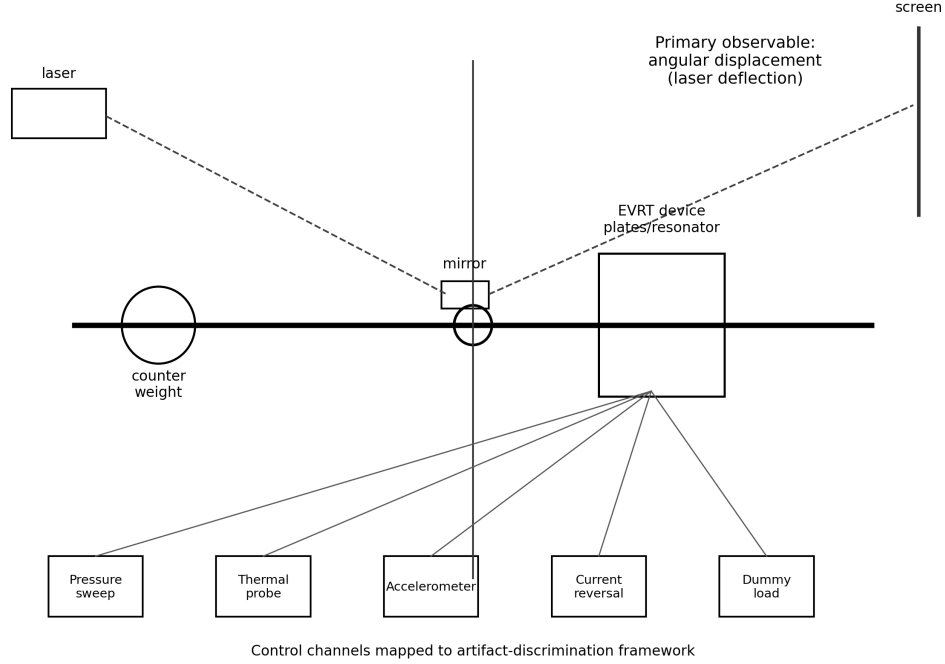


Figure 6: Experimental design figure showing a torsion-balance implementation mapped to artifact controls. The primary observable is angular displacement inferred from laser-spot deflection along the optical readout axis. The apparatus should be operated under device-on, dummy-load, polarity-reversed, geometry-reversed, and pressure-sweep conditions.

9 Interpretation Criteria

A signal should be considered a candidate residual only if it satisfies the following conditions:

1. persistence under vacuum or pressure-sweep conditions;
2. independence from measured thermal gradients and time constants;
3. independence from cable routing, grounding, and dummy-load operation;
4. survival under geometry reversal, polarity reversal, and null configuration;
5. reproducibility across independent apparatus or repeated rebuilds;
6. a reported χ_{eff} value with uncertainty and an explicitly stated normalization scale;
7. consistency of scaling behavior with the normalization relation used to compute χ_{eff} .

10 Uncertainty and Sensitivity Analysis

The effective susceptibility χ_{eff} inherits uncertainty from both the measured residual force and the normalization scale:

$$\delta\chi_{\text{eff}} = \chi_{\text{eff}} \sqrt{\left(\frac{\delta F_{\text{residual}}}{F_{\text{residual}}}\right)^2 + \left(\frac{\delta F_{\text{scale}}}{F_{\text{scale}}}\right)^2}. \quad (5)$$

Primary contributors include:

- force-sensor resolution and baseline drift;
- thermal-gradient and thermal-relaxation uncertainty;
- pressure-dependent variations;
- electrical power and phase stability;
- geometric alignment and distance tolerances.

Sensitivity analysis should be performed by varying each parameter independently to determine the dominant uncertainty channels. In this framework, an apparent residual is not meaningful unless it exceeds the combined uncertainty of the bounded artifact terms and the normalization scale used to define χ_{eff} . A residual signal is considered significant only if it exceeds the combined uncertainty envelope of all bounded artifact contributions.

11 Nonlinear Artifact Mechanisms

Many reported thrust-like signals arise from nonlinear coupling between subsystems. These mechanisms can produce net force biases even when the underlying oscillatory forces are symmetric. These nonlinear pathways are particularly relevant in oscillatory and phase-driven systems, where small asymmetries can be rectified into measurable net forces. Important examples include:

- mechanical rectification through stick-slip asymmetry;
- thermally induced expansion coupled to asymmetric supports;
- phase-lag-driven electrical imbalance;
- electromagnetic coupling to external structures or chamber hardware.

These effects can produce apparent forces of the form

$$F_{\text{apparent}} \sim A(\omega, \phi, T, \text{geometry}), \quad (6)$$

which may mimic candidate residual signals. Therefore, any interpretation of F_{residual} must explicitly demonstrate independence from these nonlinear pathways.

11.1 Example Constraint Estimate

Consider a torsion balance with a minimum detectable force

$$F_{\min} \sim 10^{-7} \text{ N}. \quad (7)$$

For an RF cavity with representative parameters

$$Q = 10^4, \quad P = 10 \text{ W}, \quad f = 1 \text{ GHz}, \quad (8)$$

the stored-energy proxy is

$$U \sim \frac{QP}{\omega} \approx 1.6 \times 10^{-5} \text{ J}. \quad (9)$$

For a characteristic length $L \sim 0.1 \text{ m}$,

$$F_{\text{scale}} \sim \frac{U}{L} \sim 1.6 \times 10^{-4} \text{ N}. \quad (10)$$

A null result at the force threshold would therefore imply

$$\chi_{\text{eff}} < \frac{F_{\min}}{F_{\text{scale}}} \sim 10^{-3}. \quad (11)$$

This demonstrates how the framework can produce quantitative constraints rather than qualitative classifications alone.

12 Conclusion

The primary contribution of this work is not the assertion of a new force, but the disciplined definition of what would remain after known mechanisms are removed. The artifact-discrimination framework presented here provides a reproducible path for evaluating historical and future electromagnetic force experiments.

Null results are interpreted as quantitative constraints on χ_{eff} , thereby establishing experimentally meaningful bounds on possible nonequilibrium electromagnetic response models. If no residual survives, the result constrains theoretical extensions. If a residual does survive, its magnitude, scaling behavior, and directional properties can be reported in a falsifiable and reproducible manner.

In this way, the framework shifts the focus from the detection of anomalous forces to the reproducible demonstration of conditions under which such forces cannot be explained by known physics.

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